

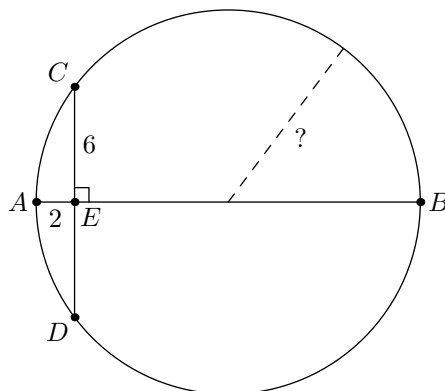
This is the Mastery Round. You have 30 minutes to complete 8 problems.
This is an individual round.

There is no penalty for guessing. Answer submissions are only valid if they are written
on the answer sheet below.

First and last name			
p	Answer	p	Answer
1		5	
2		6	
3		7	
4		8	

Problem 1. Find the least positive integer x such that $x - 1$ is divisible by 5, and $x - 5$ is divisible by 7.

Problem 2. In a circle, AB is a diameter and CD is a chord perpendicular to AB , intersecting AB at point E . Given that $CE = 6$ and $AE = 2$, find the radius of the circle.



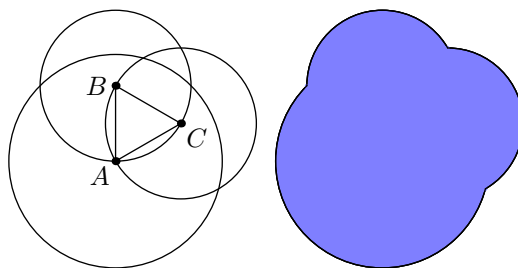
Problem 3. Austin orders two Sloppy Joe sandwiches from Bubway. Each of Bubway's sandwiches has a random size, which is a real number chosen uniformly at random between 0 and 1. After Austin gets his first sandwich, he guesses if it will be bigger or smaller than his second one. If he guesses optimally, compute the probability he is right.

Problem 4. Let $r < s < t$ be the real roots of $x^3 + 14x^2 - 34$. Compute $\frac{s}{r} + \frac{s}{t}$.

Problem 5. Let the remainder when 7^{777} is divided by 1000 be p , and the remainder when 3^{333} is divided by 100 be q . Find $p + q$.

Problem 6. Markov is climbing up a chain. The chain is 35 feet long, and Markov starts at the very bottom of the chain. Every 8 feet, Markov has a $\frac{n}{40}$ chance of falling off the chain and restarting from the bottom, where n is the number of feet up the chain Markov is. If Markov can climb at a rate of 1 foot per minute, what is the expected amount of minutes it takes Markov to climb to the top of the chain?

Problem 7. In equilateral triangle ABC with side length $\frac{\sqrt{2}}{2}$, a circle of radius 1 is drawn with center A and circles of radius $\frac{\sqrt{2}}{2}$ are drawn with centers B and C . Compute the area of the union of the 3 circles, which is shaded in the diagram below.



Problem 8. Let $a \neq 1$ be the unique positive integer for which

$$\log_3(\log_3(2025 \log_a 3^a) + 3)$$

is also an integer. Compute the number of positive divisors of a .